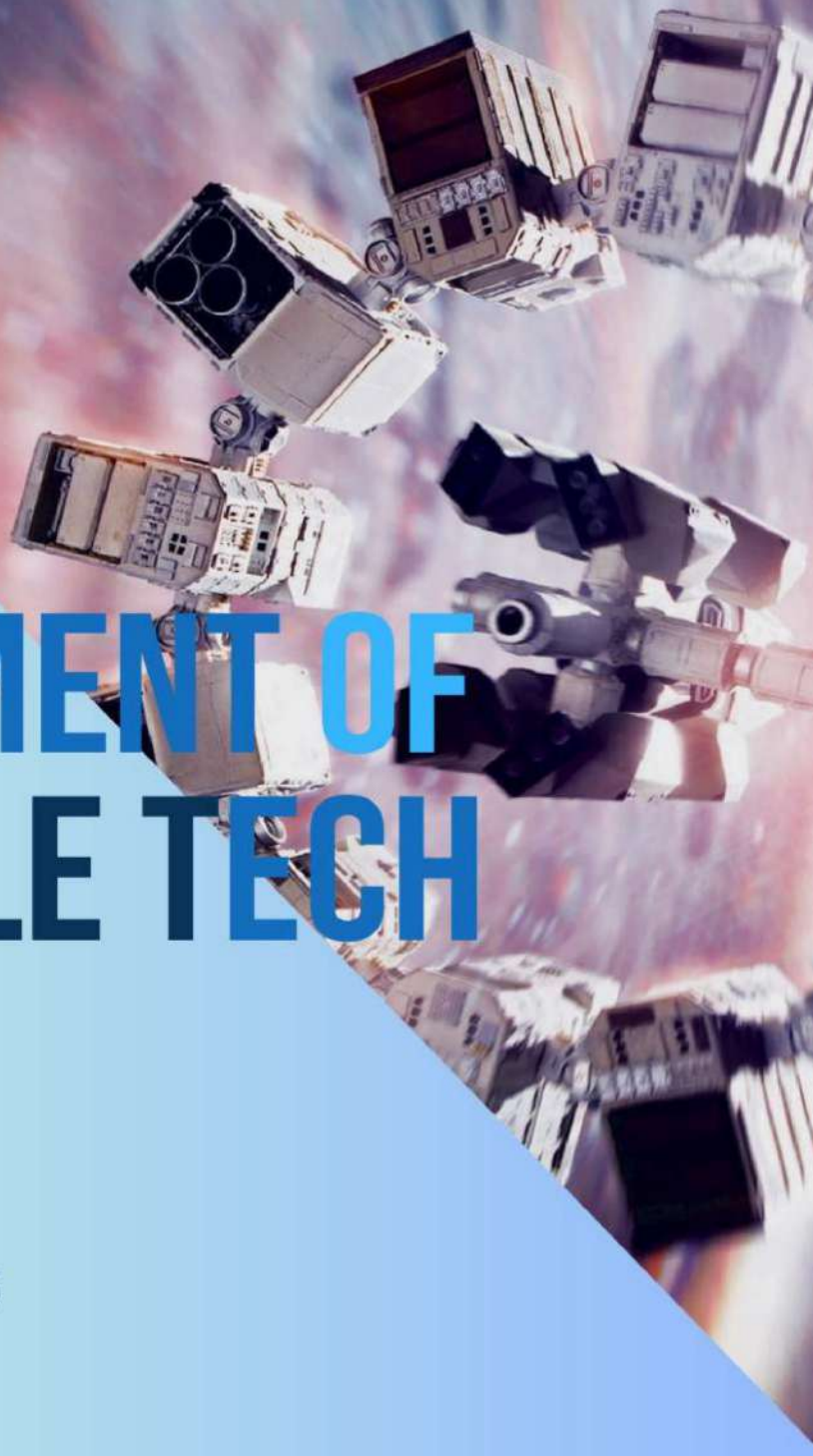
MLK Technology and Science Ltd
Presents



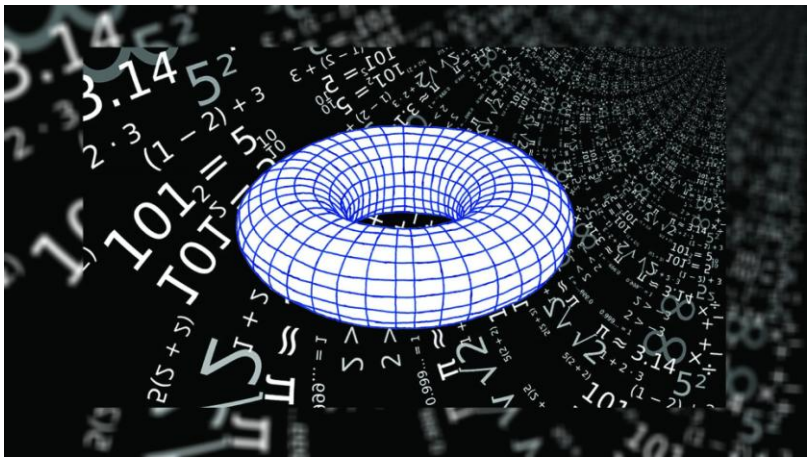
KNOW-HOW

DEVELOPMENT OF WORMHOLE TECH

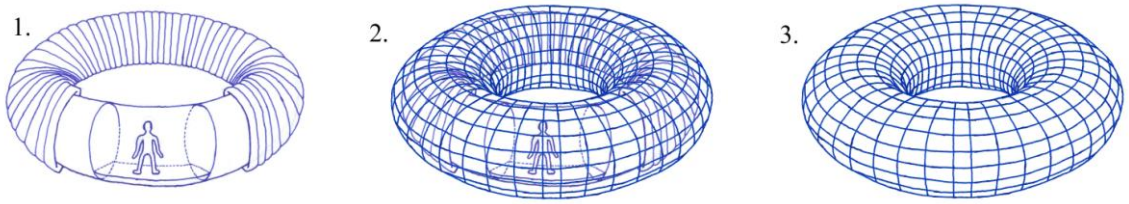
PHYSICS,
MATHEMATICS
AND ENGINEERING

CEO: Marek-Lars Kruusen
mlk@technologyandscience.eu
Tallinn, Estonia, European Union
2024

The technical concept for creating a tunnel in space-time: development of wormhole technology



The first generation technical concept for creating a tunnel in spacetime



About me and the company

"**MLK Technology and Science Ltd**" is a startup company (and business-to-business company) primarily engaged in scientific research on wormholes and technology development. To clarify, it can be said that the company is engaged in the creation of know-how. Know-how describes the physics of wormholes and their technical creation. **Company logo**:

Marek-Lars Kruusen's
technology and science

The limited company is non-profit working on developing technology and science in the field of wormholes. The official data of the company can be seen on the websites:

Link: https://ariregister.rik.ee/eng/company/17008425/MLK-Technology-and-Science-O%C3%9C

Link: https://orcid.org/0000-0002-3223-6099

The company is registered in the **Republic of Estonia** (EE, EST), which is a member state of the **European Union** (EU).

The company is also registered in the **United States of America (***https://sam.gov/content/home*):

> **NAICS Codes**: 611110, 611210, 611310, 611410, 611430, 813410, 813910, 813920. **Cage Code**: 0160J. **UEI Number**: JK29PY813VN7. **Business Name**: MLK Technology and Science Ltd. **Applicable Business Names**: MLK Technology and Science OÜ, MLK Technology and Science.

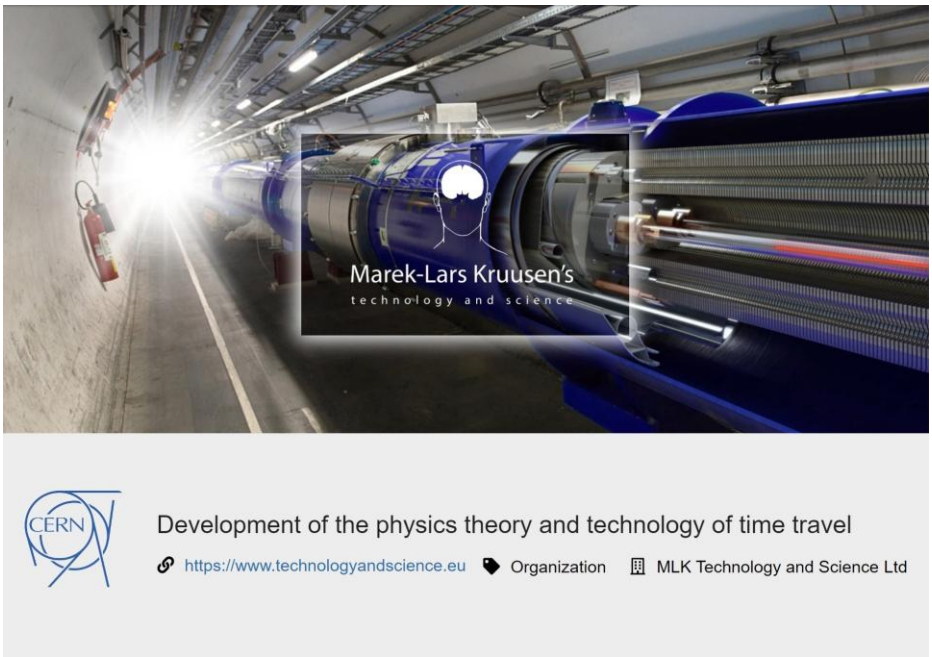
Marek-Lars Kruusen (**MLK Technology and Science**) collaborates with the **University of Cambridge**. This collaboration involves the publication of scientific papers and preprints on a platform managed by **Cambridge University Press**. For example, preprints can be found on **COE**, which is hosted at Cambridge University Press:

UNIVERSITY OF CAMBRIDGE

Link: https://drive.google.com/file/d/1pJD2WzG55ex1hVeOlwDZ03ZRR1nApjzy/view?usp=sharing

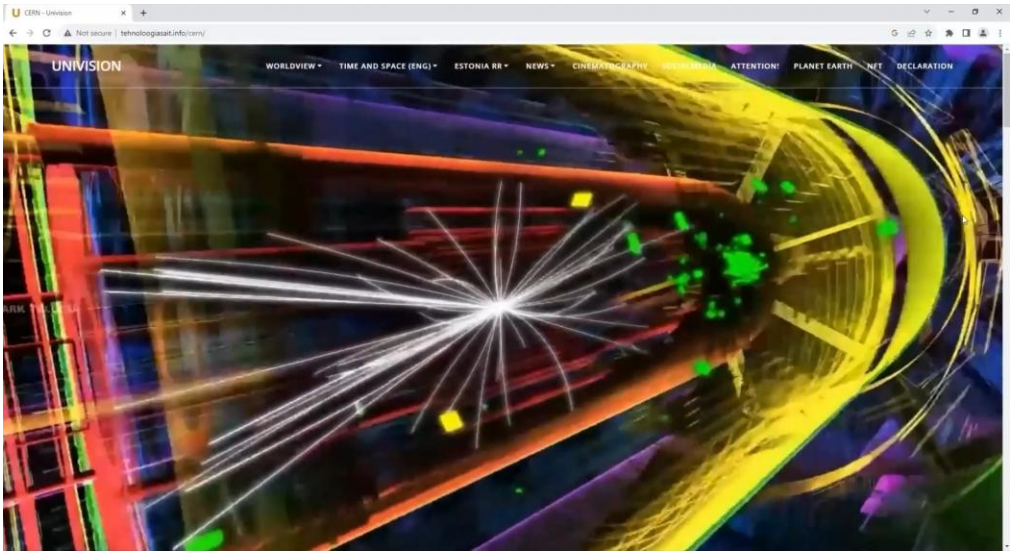
Link: https://www.cambridge.org/engage/coe/search-dashboard?authors=Marek-Lars%20Kruusen

The company "**MLK Technology and Science**" is the founder and moderator of the research department located at CERN: "**Development of the physics theory and technology of time travel**". The **European Organization for Nuclear Research**, known as **CERN**, is an intergovernmental organization that operates the largest particle physics laboratory in the world. Established in 1954, it is based in a suburb of Geneva, on the France–Switzerland border. It comprises 23 member states. Israel, admitted in 2013, is the only non-European full member. CERN is an official United Nations General Assembly observer.



See more here: https://zenodo.org/communities/time_travel/

In the file named "**CERN.mp4**," the cooperation is presented as a video:



Link: https://drive.google.com/file/d/196WJD5OS-ii0XeDAJiwRqyhEQqqghaEy/view?usp=sharing

Homepage: https://www.technologyandscience.eu/cern/

I have studied higher physics and mathematics at **Tallinn University**, which is located in the **Republic of Estonia**. I have acquired knowledge concerning information technology at **Tallinn Polytechnic School** and mechatronics and robotics at **Tallinn Industrial Education Center**. Studies concerning physics and mathematics have been carried out for approximately 17 years, the last 7 of which have been especially fast and fruitful. All the information is available on a webpage I created (**homepage**):

Company homepage: https://www.technologyandscience.eu

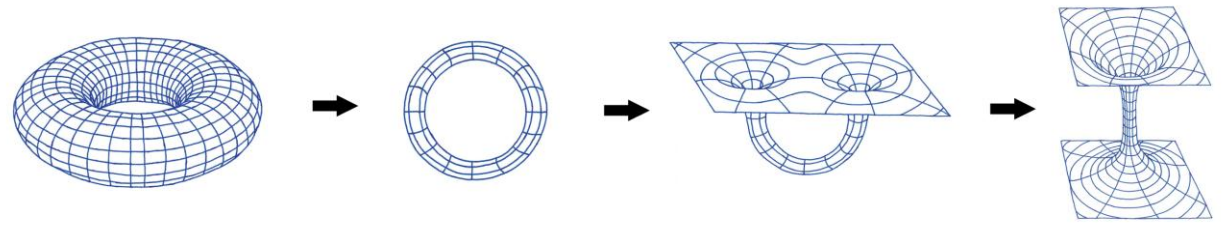
In the file named "**Front Page - Univision.mp4**," the company's website is presented as a video:

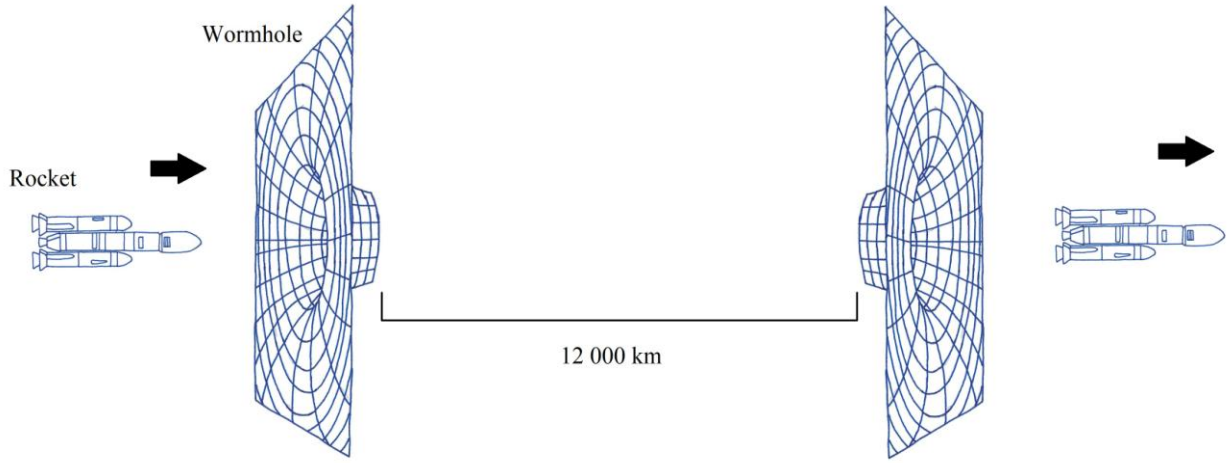
Link: https://drive.google.com/file/d/1m10lpUTSOzZxSvR7Lu7xN7SekEd9Bne3/view?usp=sharing

All the work and articles that I have ever published have been presented there. Some of the information is presented in Estonian and some of the information is presented in English. A lot of translating has to be done from Estonian to English.

What are wormholes and can they really be created?

A tunnel in spacetime or a wormhole is a curvature of spacetime connecting two points in spacetime, which could allow moving from one point in time to another or moving from one point in space to another in an instant:





A tunnel in spacetime can be created with energy. However, the problem lies in the fact that a tunnel in spacetime penetrable for human beings can only be created with excessive energy, which we cannot get from anywhere. This is where the majority of difficulties lies in creating them. For example, in order to create an object the same size as a tunnel in spacetime penetrable for human beings, much more energy would be needed than the energy corresponding to the total mass of the Earth.

> For example, in 2019 the energy production/usage of the entire world was 158 800 TWh or 158 800 000 000 000 kWh. Since in physics kilowatt hour is a multiplied unit of watt second, we can present the size of such energy in joules: $E = 5{,}7168*10^{20}$ J. However, during the years 1800 – 2019, the whole energy production/usage was approximately 7 800 000 TWh or 7 800 000 000 000 000 kWh, which equals $E = 2{,}808*10^{22}$ J in joules. Such amount of energy is not enough to create a proper wormhole, since approximately energy in the magnitude of 10^{44} J would be required to create a human-sized tunnel in spacetime, but it is not possible to take it from anywhere or produce it. Since quantities of energy corresponding to the mass of planets cannot be produced or used, the only remaining possibility is to use minimal energies, which is clearly already within the limits of current technical capabilities of mankind.

See more here:
https://drive.google.com/file/d/1TlywakyTyq1mpHoKMkCU8x7rbYpq1Xlx/view?usp=sharing

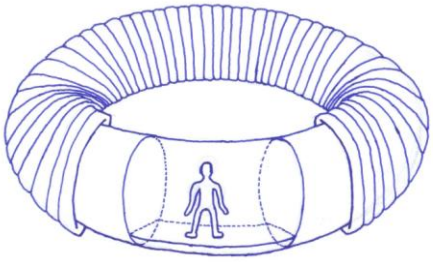
The solution can be found in the speed of generating or changing of an electromagnetic field, rather than the strength of the energy of the field. There lies the novel solution I propose. High energy bends spacetime, but the creation or changing of an energy field with the speed of light also bends spacetime.

The physical and mathematical interpretation of wormholes has actually been known for about a hundred years, but their technical creation has not been possible until now. However, in this work, apart from the nature of the wormhole, its technical feasibility is also shown, which has not been presented before. One of the biggest obstacles to the technical creation of wormholes was the production of the necessary energy, but the new discoveries presented in this work show that this is not necessary. The key to the creation of wormholes lies in the changes or creation of energy fields that occur exactly at the speed of light. Physical states that change or arise at the speed of light are also accompanied by the emergence of abstract geometric surfaces with time and space curved to infinity.

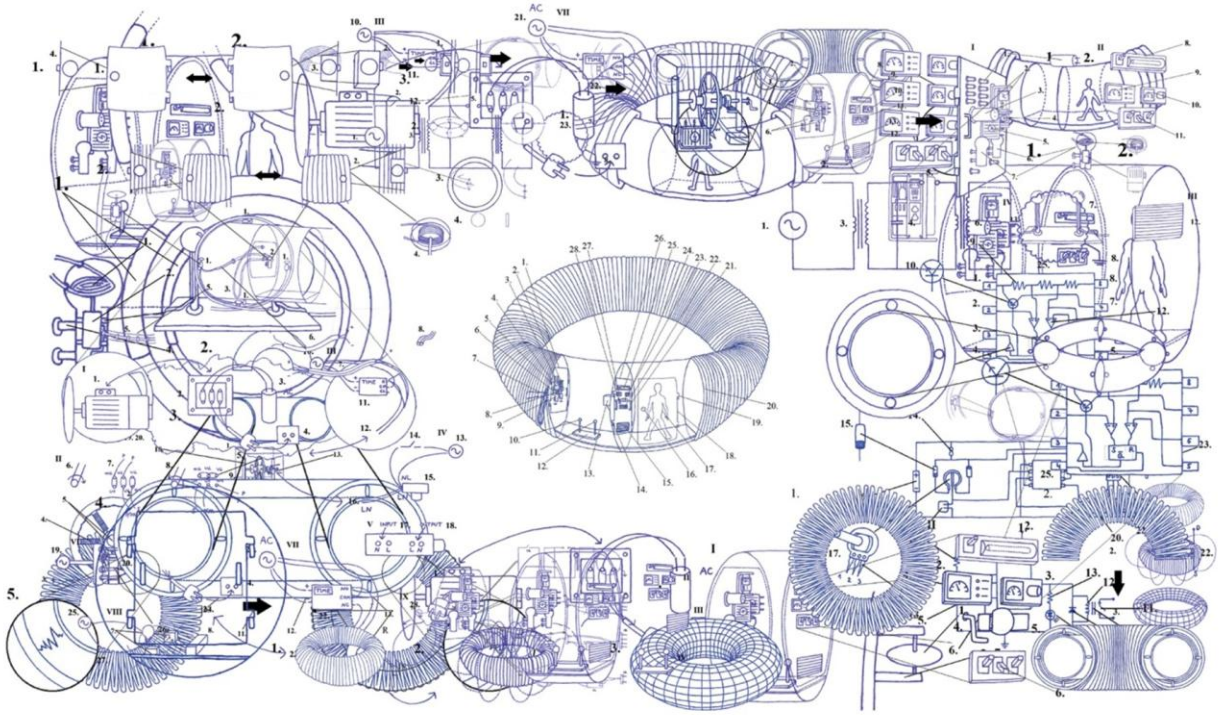
The actual creation of wormholes is crucial to humanity's plans for the future. They would be practically indispensable for, for example, space travel to great distances, and wormholes would also allow people to travel in time to the past and the future. Time travel could be used in the study of human history and would be a very good tool for accurate weather forecasting, for example. For a long time, wormholes were considered the realm of science fiction and fantasy literature, and in some cases even pseudoscience. However, new discoveries concerning the possibility of their creation, which are thoroughly presented in this work, make the existence of wormholes a tangible reality.

My project

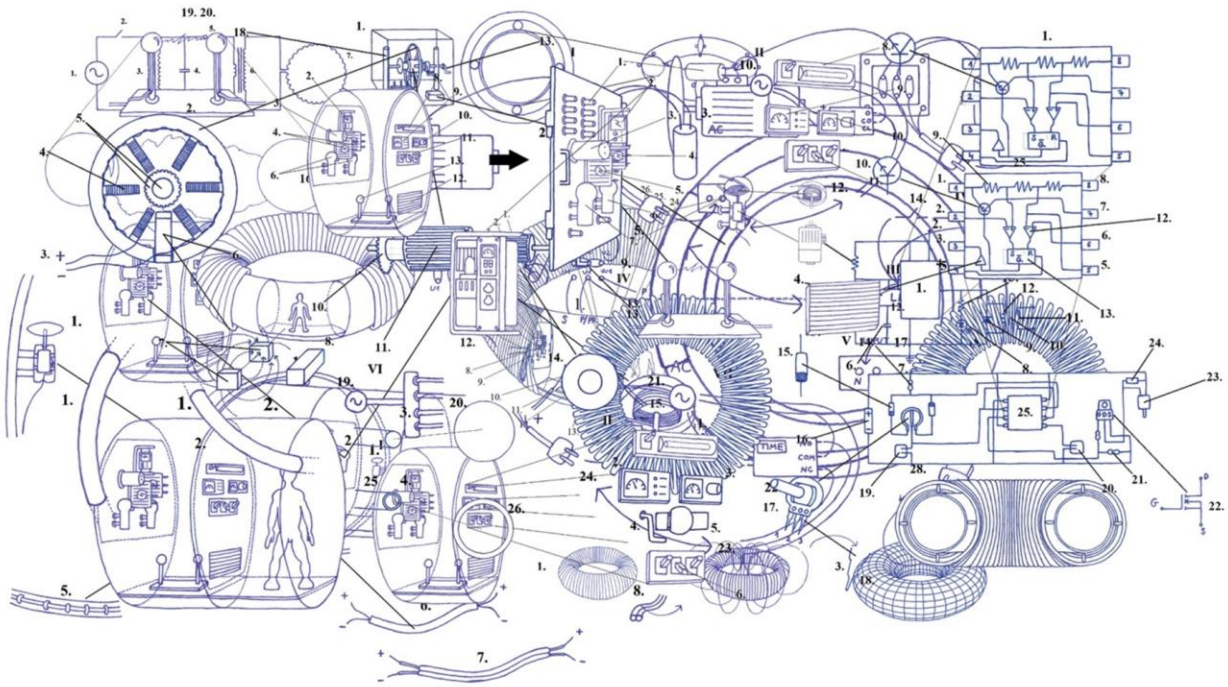
The invention that I describe would make it possible to artificially create tunnels in spacetime. A loop-shaped tunnel in spacetime is created by a rotating ring machine, which has it's own machinery space and control room as well as electromechanical components creating the electromagnetic field. It is a human-sized rotating machine:



The device/machine consists of a large number of elements of the circuit, all of which are closely related to each other. This is shown in the following figure, in the center of which you can see the compact machine and, around it, the elements of the circuit that make up the machine:

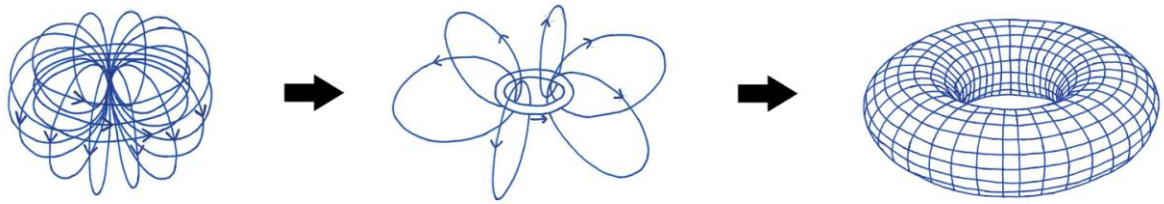


In order to save space, all elements of the machine are shown in one diagram:



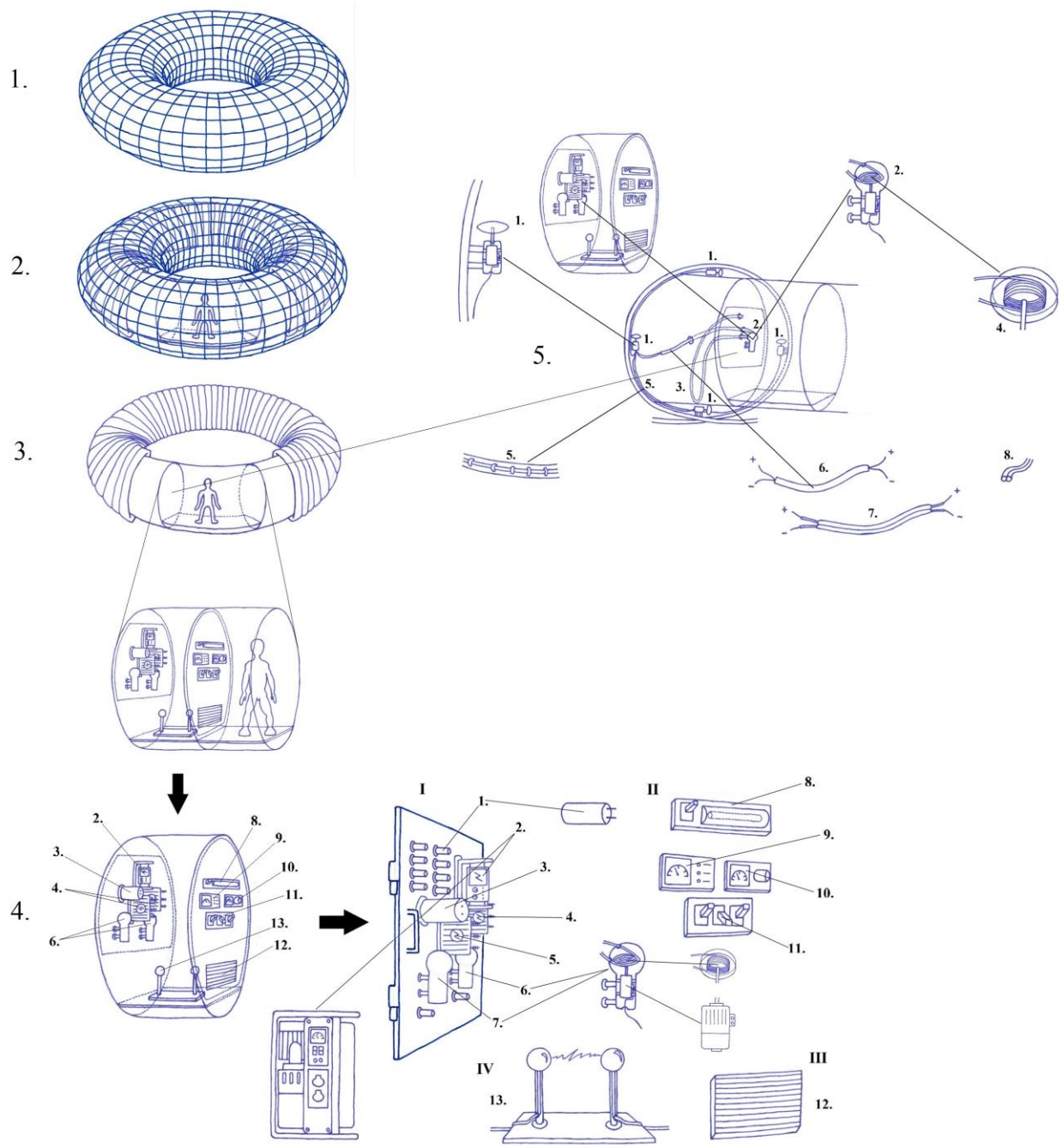
In "Materials", we will present and examine each element of the machine's scheme in more detail.

The ring machine allows artificially creating a loop-shaped tunnel in spacetime, which in turn allows a person to teleport in time and space. A ring machine consists of a toroid and a tube inside it that must rotate. There is a machine and control room inside the pipe, where a person can enter. There are electromechanical components that create an electromagnetic field around the toroid and that cause the tube to move or rotate. The tube is rotated by 8 wheels, all of which are equipped with electric motors. The entire machine is controlled by switches. The main purpose of the ring machine is to create an electromagnetic field around the machine that changes/is created exactly at the speed of light c.

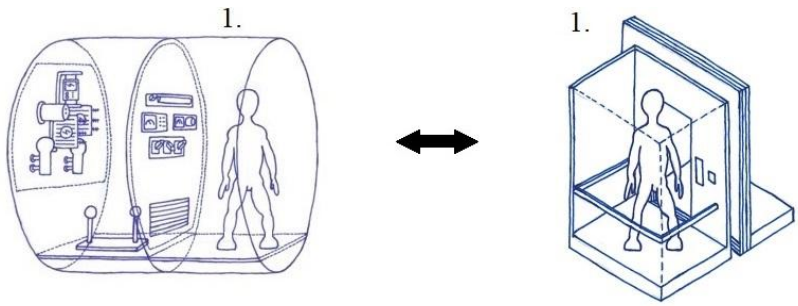


According to Albert Einstein's theory of relativity and Marek-Lars Kruusen's physics theory of time travel, the electromagnetic field changing/emerging at the speed of light c should create a closed trapped surface in spacetime, which can also be understood as a tunnel in spacetime. The definition of trapped surface in spacetime is known precisely from the physics of black holes, where infinite transformations of time and space occur.

General drawing of ring machine:

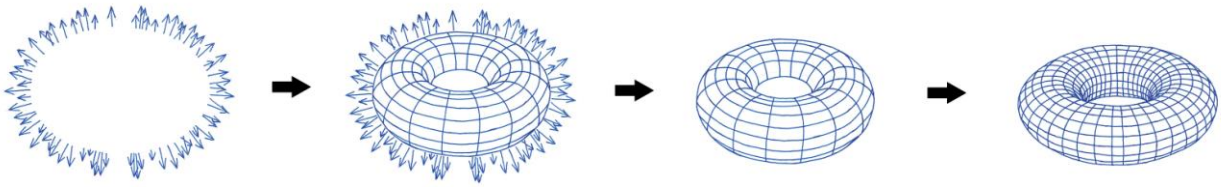


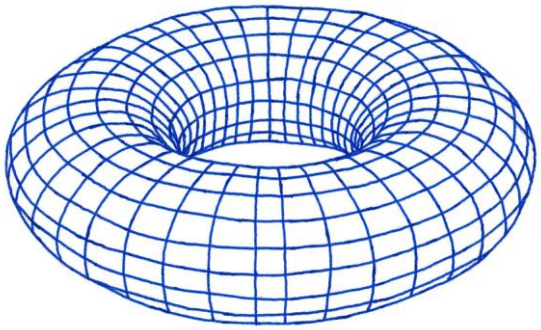
Such an invention is similar to an elevator used in buildings, as in both cases a person enters an enclosed space (1):



which is transported to another location using wheels and wires (cables, ropes). However, the difference is that while an elevator carries a person only vertically up or down in the building, the ring machine transports the person in the toroid along a ring-shaped tube back to the original location. In this sense, the invention can also be used as a ring-shaped elevator that would transport people from one location to another, but in such a way that the final destination coincides with the initial destination.

The aim of the invention is to artificially create a loop-shaped tunnel in spacetime:





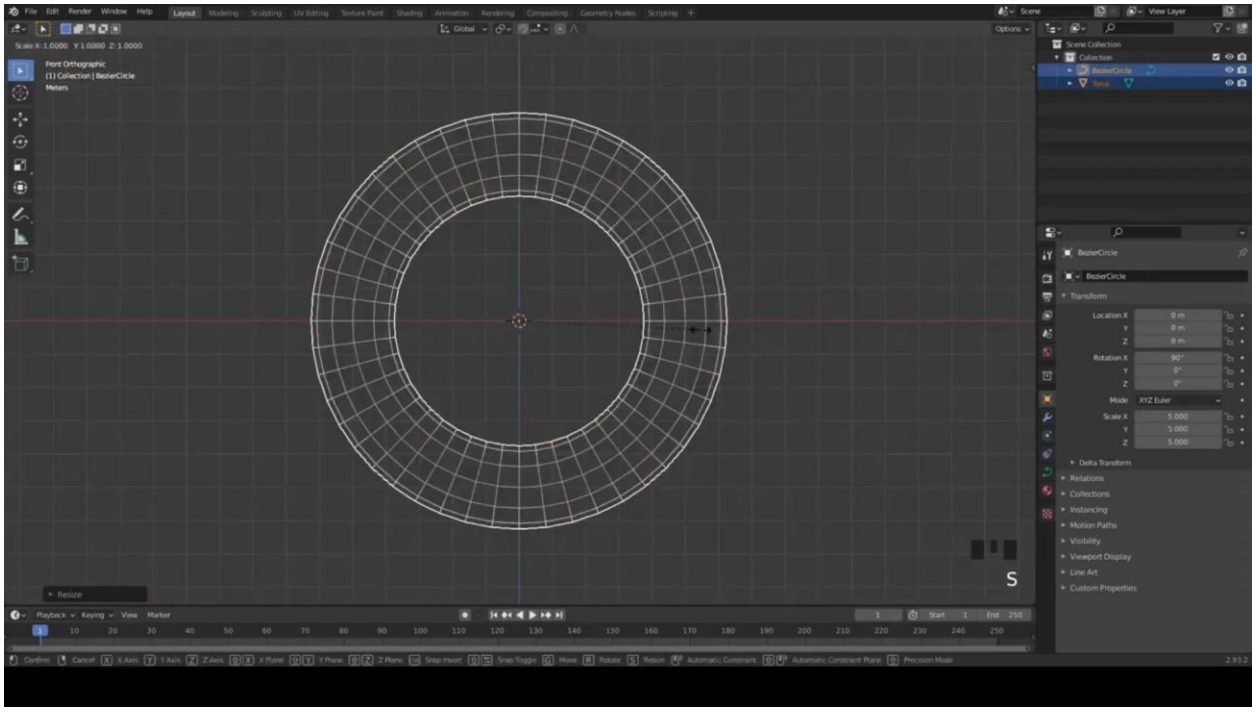
This is achieved with a solution which is thoroughly described in the study that I composed. My proposal is that the engineers and scientists could finish off the technology I propose. Technology concerning tunnels in spacetime could replace rocket technology in the future. I have evidence-based physical and mathematical science and an early stage technical model for creating a tunnel in spacetime. Developing this specific technology will create new opportunities to explore human history and also to move in space.

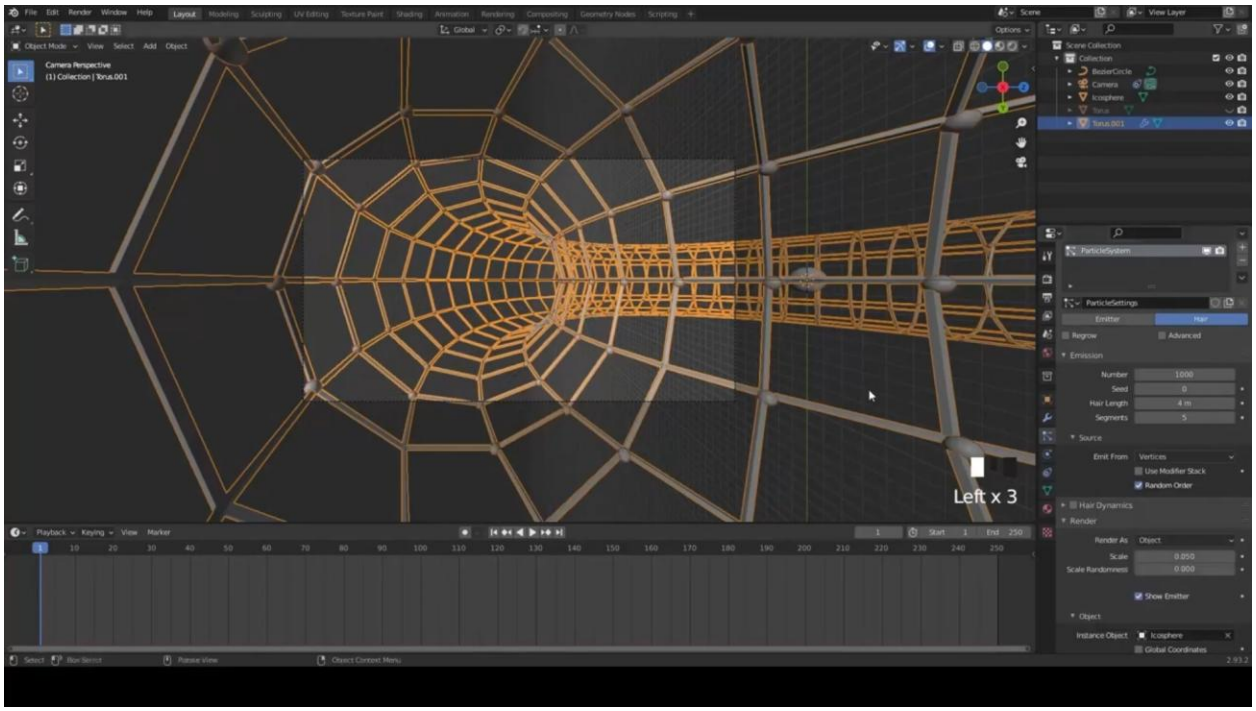
The annular tunnel in spacetime (3D)

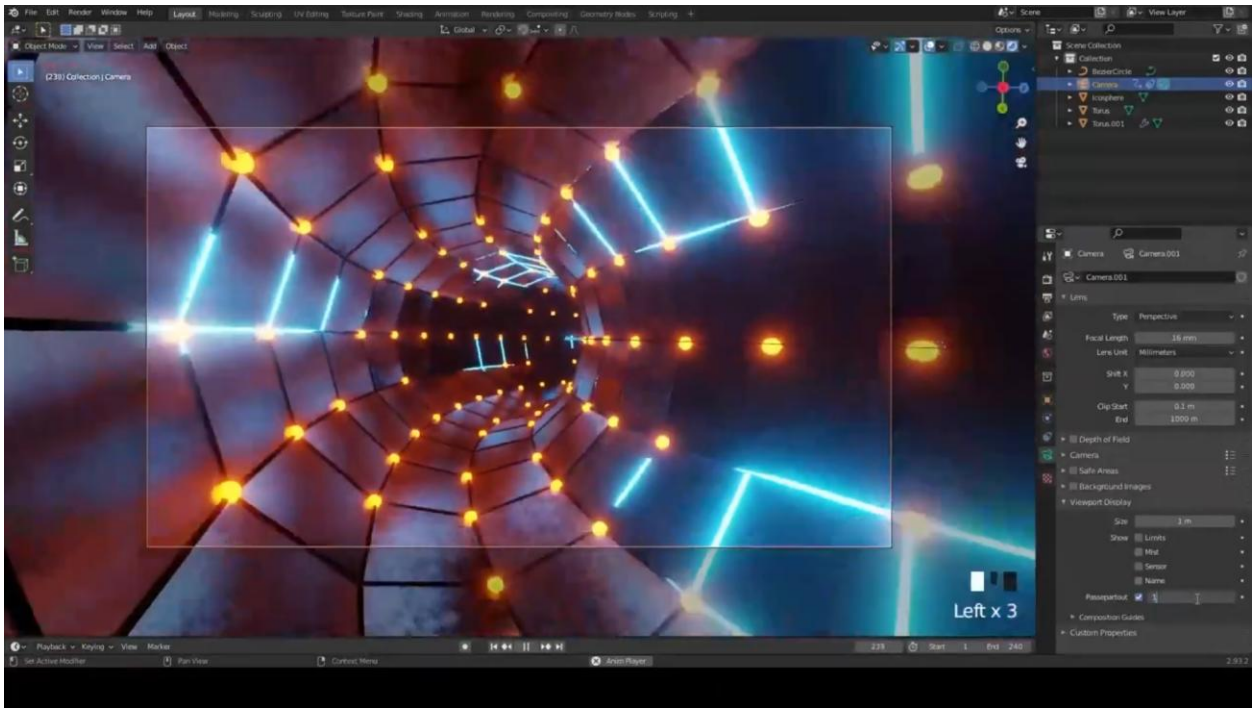
The annular tunnel in spacetime has been visualized in a computer program called **Blender**. Blender is a **computer program** that can be used to create animated three-dimensional objects and environments. A computer program called **3D Blender** can be downloaded and used for free at the Internet address: *https://www.blender.org/*. A three-dimensionally visualized, or animated, annular tunnel in spacetime creation video can be viewed and followed on the **network drive**:

Link: https://drive.google.com/file/d/1EdlLQPdzNpKzAQUff0jmUamgkmxngWb7/view?usp=sharing

Below is a whole series of **screenshots** (pictures) from the previously mentioned video:







Harvard University: https://doi.org/10.7910/DVN/XNGHKB

The uniqueness of my project

The technical solution for creating tunnels in spacetime presented in this work differs from the technologies on the market in that the ring machine creates a ring-shaped tunnel in spacetime, i.e. a ring-shaped wormhole, **with much simpler technical means and does not consume large amounts of energy**. This is one of the main differences - the product does not consume large amounts of energy compared to devices used in rocket technology. The energy consumption of a production-ready machine is many times lower compared to other products in the space technology sector, **which is why it is also much "greener"**.

Another big difference is that the ring machine presented in this work is completely unique. This means that there is no product on the market that would allow artificially creating a tunnel in spacetime or a wormhole. Not even NASA or private space companies have this kind of technical concept described in this work. **It is the world's first technological concept to create a tunnel in spacetime** that a person can pass through and that would be technically controllable.

Any technology that creates a wormhole is extremely necessary for the market, as it allows **space travel to be made much easier, economically cheaper and also "greener"**. As the space industry is on the rise all over the world, so any company would benefit greatly from the space industry. Any company could become a part of the space industry sector, as **billions of US dollars** move in this economic sector every year.

Machines that generate wormholes would replace rockets and other spacecraft in the future. Therefore, wormhole technology (through it also the company) would have immense value in the future - valued in the billions of dollars.

The actual creation of **wormholes** is crucial to humanity's plans for the future. They would be practically indispensable for, for example, **space travel** to great distances, and wormholes would also allow people to **travel in time** to the past and the future. Time travel could be used in the **study of human history** and would be a very good tool for accurate **weather forecasting**, for example.

Contents of my project

The technology that creates ring-shaped tunnel in spacetime would quite obviously be technically useful for space industry. This has been depicted in many science fiction movies and fictional fantasy novels. A tunnel in spacetime or a wormhole enables moving very long distances in space in a very short period of time. This possibility is principally irreplaceable for

space travel, compared to which rocket technology will be left in the dustbin of history. Therefore, I have no direct need for a client support letter. For me, the main purpose of development of theoretical physics and practical engineering is to reach such a level of technology that it will be possible to create a tunnel in spacetime which an actual human being could pass, use it safely and control it.

In technology industry, scientists and engineers could create and use my proposed technology or a similar technical solution. Development plan would logically and briefly be the following:

- Creation and development of physics and mathematics describing tunnels in spacetime and presenting it to investors.

- The progress in physical sciences and mathematics is the basis of technical feasibility of wormholes. Understanding this is essential for any space agency.

- Creating, testing and developing engineering (for example mechanical engineering and electrical engineering) in the field of wormholes.

- The final and successful technical implementation or demonstration could also be carried out in the laboratories.

To accomplish the stages listed above, financial support is needed. I'll be direct: I need money to hire a team, purchase equipment, procure materials, rent spaces, develop technology, refine many details, conduct experiments, develop technology and expertise, develop new generation models, etc. Therefore, I am reaching out to investors and supporters.

In the long term, I aim to establish a new technology industry sector dedicated to the development and advancement of wormholes. Wormholes would be entirely different transportation enabling entities. This has not been done in the world yet. It requires significant investments and support from backers.

The entire implementation as well as proof of technical conception has been presented in a one network drive, which has been linked below ("**Materials**"). The final and successful demonstration of the technical implementation could be carried out in the laboratories of industry. In the future, the user of the technology that I propose could in principal be space agencies.

Patent application in the United State of America

The invention, which allows creating a human-sized annular tunnel in space-time, has been attempted to be patented in the **United States Patent and Trademark Office** (**USPTO**, website: *https://www.uspto.gov*). The corresponding patent applications by Marek-Lars Kruusen have been filed in 2023. Patent application number is: **18/339,426**; and title of invention: „**Method and equipment for creating a tunnel in spacetime or wormhole**“.

However, only the engineering-technical part is presented in the patent application, which should convincingly demonstrate the creation of a tunnel in spacetime, and that's it. According to the **Patent Act**, only engineering and technical solutions are patentable, not physical theories. Therefore, the theoretical physics part of how a tunnel in spacetime causes a person to teleport in time and space remains unreported in the patent application. This is already a theoretical part, which is not a patentable object according to the patent law.

Link: https://drive.google.com/file/d/1IltwwjKpmACcoWUuB5Ypc0T0jf0Nqmv5/view?usp=sharing

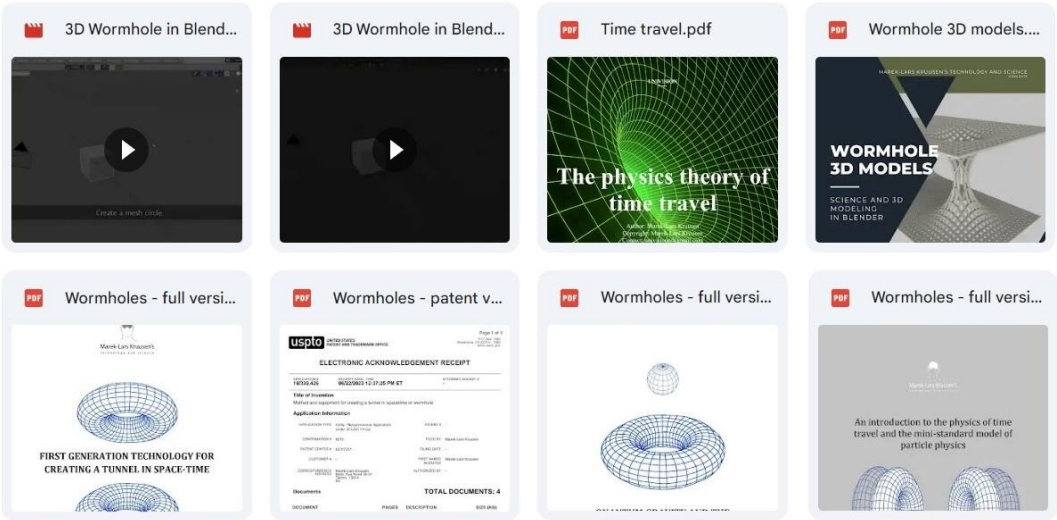
Team, profitability and time

I offer engineers the knowledge and technical concept to create a wormhole. MLK Technology and Science company can assemble the team required to create the wormhole themselves. I would prefer to be in a consulting role. Company can also determine the budget required for the implementation of the wormhole technical concept. Space agencies may use my proposed technical concept at any time, but after payment of the fee I have specified.

The development of physics and mathematics has taken years. Forming a team of engineers and scientists, developing the technology, and conducting experiments would take about a year. Such a timeframe must be considered, as the work is very extensive and highly complex.

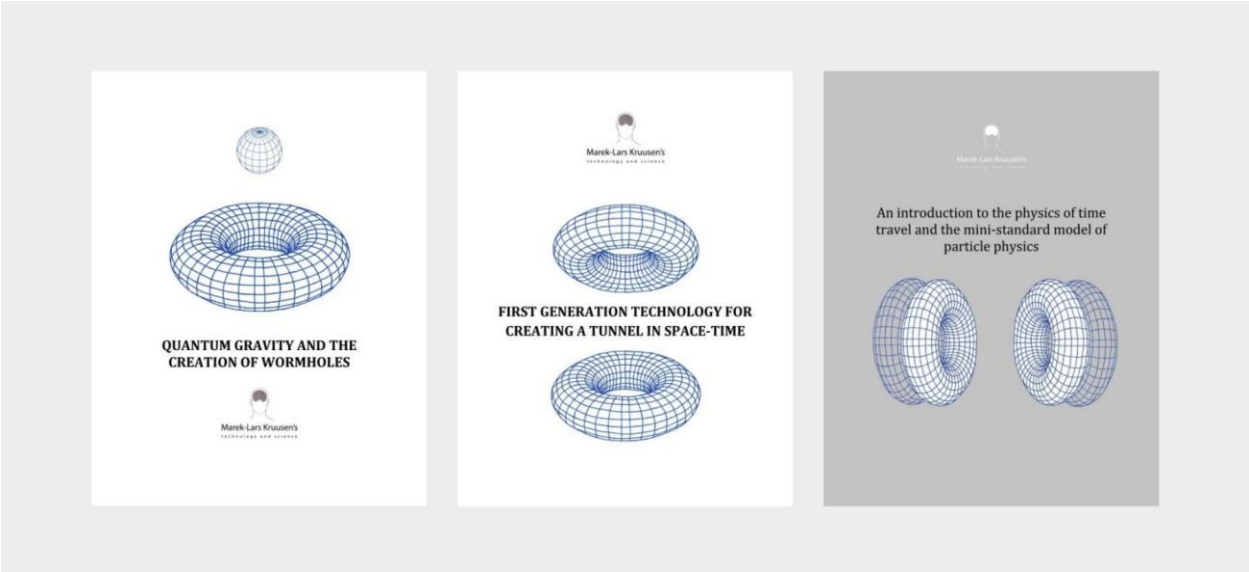
Materials

Materials describing the physics and mathematics of wormholes and their creation technology can be found on the **network drive**. This can be shared with **everyone** through the following link:



MLK Technology and Science: https://drive.google.com/drive/folders/1iPD9S-MLJmSiQMeAHwi616VlGX6ZjX3A?usp=sharing

THE MOST IMPORTANT: the PDF files named "**Wormholes – full version.pdf**", "**Wormholes - full version 2.pdf**" and "**Wormholes - full versions 3.pdf**" contains the physics and mathematics of wormholes and their creation through electrical engineering:



The patent material is presented in the PDF file named "**Wormholes – patent version.pdf**," which has been submitted to the **United States Patent and Trademark Office (USPTO)**. It is currently under review.

In the file named "**Wormholes – patent version.pdf**," there is also a document confirming the application submitted to the **USPTO**:

uspto UNITED STATES PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/339,426	**06/22/2023 12:37:25 PM ET**	**-**

Title of Invention

Method and equipment for creating a tunnel in spacetime or wormhole

Link: https://drive.google.com/file/d/1k9qDU89OPn6LV3ve03AoHqu_1KWffd2z/view?usp=sharing

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Marek-Lars Kruusen collaborates with the **European Organization for Nuclear Research** (CERN). This collaboration involves the publication of scientific papers and preprints

on a platform managed by CERN. In the file named "**CERN.mp4**," the cooperation is presented as a video.

Risks and challenges

This project comes with certain risks and challenges, which are actually common for new and innovative technical solutions. One risk is whether the given technical concept actually works in reality. Its reliability lies in the sciences of physics and mathematics. Theoretical physics and mathematics convincingly demonstrate the possibility of wormhole formation through the change in electromagnetic field speed. There is no reason to doubt this. However, the technical solution is vast and complex. To ascertain its functionality, experiments are necessary. My scenario envisions several generations of technical models; the current solution is just the first.

If the first model doesn't work as intended, a second-generation model will be built, taking into account the flaws and oversights of the first generation. In theoretical physics and mathematics, there's no reason to doubt, but technical concepts need to be verified in reality. History has shown that if theoretical science is very strong, sooner or later, it can also be implemented technically. For example, Elon Musk failed many times before creating a reusable rocket that works in reality. I must admit that I must also be prepared for such a scenario: prepared for failure. However, unfortunately, this is inevitable for success. Success consists of failures.

The development of physics theories forms the basis for the realization of any technology. The credibility of this project can be verified by examining the materials that I share as links. Research has been conducted for over 10 years and it includes hundreds of pages of physics and mathematical analysis. Everything is publicly available and open for examination.

Long-lasting design

The lifespan of the ring machine could be approximately the same as, for example, a car. Since the machine is made of durable material (iron or metal), the machine itself is therefore also durable over time. I can confirm that the machine design is long-lasting, as it includes a construction of iron or steel, which is durable over time. The utilization of wormhole physics is essential for the development of wormhole technology. Knowledge of physics and mathematics never becomes outdated; rather, it evolves over time.

Reusability and recyclability

I can confirm that the machine described in this project is reusable and recyclable. Recyclable in the sense that its parts can be repaired, exchanged, and used in other technology industries, such as the transportation sector. The ring machine can be used multiple times. It is not a single-use technology. For example, the elevator used in buildings is a reusable machine. Similarly, the ring machine is also a reusable machine.

Sustainable materials

The machine contains iron or metallic compounds, as the structure must be very strong. The machine can be built with sustainable materials. It is important that some parts of the machine conduct electricity, and the machine's supporting structure must also be strong: moving parts must function smoothly.

Environmentally friendly factories

The factory must be able to produce machines with a production cycle that has minimal impact on the environment. It must not leave too much production waste or toxic chemicals that are harmful to the environment. The factory must comply with internationally recognized standards regarding environmental sustainability impacts. The eco-friendliness of factories depends on which factory will start producing my machines. I can also make the choice myself: if the factory is not eco-friendly, I can refuse its services and try to choose a more environmentally sustainable factory. The ring machine creates a ring-shaped tunnel in spacetime, i.e. a ring-shaped wormhole, with much simpler technical means and does not consume large amounts of energy. The product does not consume large amounts of energy compared to devices used in rocket technology. The energy consumption of a production-ready machine is many times lower compared to other products in the space technology sector, which is why it is also much “greener”.

Sustainable Distribution

The space industry would benefit the most from my invention. Since space-faring nations are major industrial powers, they have large factories of their own where my technology could be

manufactured. This means that there wouldn't necessarily need to be extensive international transportation. Instead, space-faring nations would produce them domestically. I don't foresee any supply issues within countries. Sustainable implementation and dissemination depend on the demand for machine consumption. The primary consumers are the transportation sector and the space industry. NASA and ESA, as well as private space companies like SpaceX, would certainly be interested in the development of the technology and science described in this project and in collaboration.

Something else

Since the manufacturer of the ring machine technology is "MLK Technology and Science," the entrepreneur hires an environmental inspector who assesses, observes, and advises on making the technology's production and delivery more environmentally friendly. This would likely be the most effective solution in this field. The company's manager is not an expert in environmental sustainability. Therefore, it is necessary to hire experts who conduct various inspections.

Frequently Asked Questions

What is the name of the company?

According to the documents of the Republic of Estonia, the name of the company is: MLK Technology and Science OÜ. However, the internationally available name is: MLK Technology and Science Ltd, formerly known as "Marek-Lars Kruusen's technology and science". The acronym "MLK" stands for the name: Marek-Lars Kruusen.

Is the topic of wormholes fantasy or reality?

The physical and mathematical interpretation of wormholes has actually been known for about a hundred years, but their technical creation has not been possible until now. However, in this work, apart from the nature of the wormhole, its technical feasibility is also shown, which has not been presented before. One of the biggest obstacles to the technical creation of wormholes was the production of the necessary energy, but the new discoveries presented in this work show that this is not necessary. The key to the creation of wormholes lies in the changes or creation of energy fields that occur exactly at the speed of light. Physical states that change or arise at the speed of light are also accompanied by the emergence of abstract geometric surfaces with time and space curved to infinity. The actual creation of wormholes is crucial to humanity's plans for the future. They would be practically indispensable for, for example, space travel to great distances, and wormholes would also allow people to travel in time to the past and the future. Time travel could be used in the study of human history and would be a very good tool for

accurate weather forecasting, for example. For a long time, wormholes were considered the realm of science fiction and fantasy literature, and in some cases even pseudoscience. However, new discoveries concerning the possibility of their creation, which are thoroughly presented in this work, make the existence of wormholes a tangible reality.

Is wormhole technology patented?

The patent application was submitted in June 2023 to the United States Patent and Trademark Office. It is currently under review. According to estimates, the office should issue the patent in December of this year.

What is the purpose of wormhole technology?

It is the world's first technological concept to create a tunnel in spacetime that a person can pass through and that would be technically controllable. Any technology that creates a wormhole is extremely necessary for the market, as it allows space travel to be made much easier, economically cheaper and also “greener”. The actual creation of wormholes is crucial to humanity's plans for the future. They would be practically indispensable for, for example, space travel to great distances, and wormholes would also allow people to travel in time to the past and the future. Time travel could be used in the study of human history and would be a very good tool for accurate weather forecasting, for example.

In what sense is the ring machine "green" or environmentally friendly?

The ring machine creates a ring-shaped tunnel in spacetime, i.e. a ring-shaped wormhole, with much simpler technical means and does not consume large amounts of energy. The product does not consume large amounts of energy compared to devices used in rocket technology. The energy consumption of a production-ready machine is many times lower compared to other products in the space technology sector, which is why it is also much “greener”.